BrickPlan: A Benchmark for Visual Assembly and Error Recovery

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Figure 1. **One benchmark, complementary assembly settings.** Examples from the complete current collection: source-derived static configurations (S), complete objects and mechanism structures for dynamic assembly (D), and community-designed modules (M). The original chair, table, car, and crane remain in D. In total there are 5,260 current case records: 3,340 static and 1,920 dynamic. The supplement lists every configuration; older versions are not counted again.

Abstract

Visual assembly requires an agent to identify structural errors, execute legal placements, and recover when earlier work changes. We introduce BRICKPLAN*, a benchmark for static repair and continuous assembly under hidden interventions. Its current collection contains 5,260 case records across three complementary tracks: 3,292 static questions on 160 source-derived configurations; 1,872 dynamic episodes on 42 complete objects and 26 mechanism structures; and 96 static/dynamic cases on twelve community modules with explicit remaining-part manuals. Earlier objects and tasks are retained within these tracks. Agents observe actual scene states and manipulate one brick at a time; success requires legal execution and the correct terminal geometry rather than one prescribed build order. A deterministic study on two legal routes per community module finds that exact-reference triggers cover only 9/36 alternative-route fault trials, versus 36/36 for actual-state interventions. Geometry replays establish solvability, while observation audits expose distinct visibility and action-cancellation challenges. We specify source-separated model comparisons, independent human operation, error-wise analysis, and observation ablations, with explicitly unmeasured result tables. The reported results are executor diagnostics; formal agent and human performance remains to be collected.*

1. Introduction

Consider a nearly completed toy car. An agent installs the left side of the cabin before the right side; another agent reverses that order. Both may be correct. If an evaluator expects the first sequence, a legitimate intermediate state can be mistaken for an assembly error. A related problem arises when a hidden fault is scheduled only after the agent reproduces one reference prefix: a correct alternative route may never encounter the fault at all. Final success, exposure to a fault, and adherence to a demonstration are different properties.

The visual specification creates a second ambiguity. A single final image may conceal joints or internal parts, making reconstruction underdetermined even for a person. A failed reconstruction then says little about error monitoring. Our community-module track supplies the surrounding object, a fixed goal, and an accessible manual for a small re-

maining module. The manual specifies the pieces and their intended assembly, but serves as a reference rather than a compulsory trajectory. Figure 1 situates these modules within the full benchmark.

The benchmark retains the previously developed static configurations, complete objects, and controlled structural mechanisms. The twelve community modules extend this collection with source-preserving context and manuals; they do not replace the earlier tasks. Static repair tests localization and correction from a supplied state, while the dynamic tracks additionally test progress monitoring and recovery during construction. Together they cover furniture, vehicles, machines, household objects, buildings, and source-derived subassemblies.

We study two complementary tasks. In *static repair*, the agent receives a faulty state and an explicit acceptance target. In *dynamic completion*, it builds a complete object or a remaining module, observes an unannounced change to previous work, and continues toward the original target without being restarted. Both expose one-brick attach and detach actions, uniform inspection tools, and fresh visual observations. Their shared principle is that accepted operations must be executable and the terminal object must be correct; intermediate agreement with a chosen reference sequence is not a success criterion.

Prior work provides important components of this setting: physically checked generation [1, 5], visual manual execution [10], single-step error correction [9], and interactive structural understanding [8]. Recent agentic design and robot-recovery benchmarks further broaden the evaluation of construction and recovery [2, 3]. Our focus is the evaluation contract connecting a visual history, legal alternative assembly orders, and hidden changes within a continuing episode.

Contributions. We make three contributions. (1) We define a visual assembly and repair protocol that accepts equivalent terminal structures, selects feasible interventions from the actual state, and separates completion from intervention exposure. (2) We assemble a three-track benchmark with 5,260 current cases, retaining 160 static configurations, 68 dynamic structures, and twelve community modules, with source and version relationships made explicit. (3) We provide reproducible route-sensitivity diagnostics, reference replays, physical-action rejection checks, and image provenance. Our present experiments establish properties of the benchmark and executor. They do not establish a population-level ranking of visual agents or independently validated human performance.

2. Related Work

Generating and understanding brick assemblies. BrickGPT generates buildable structures with validity filtering and physics-aware rollback [5]. BrickNet represents a larger part vocabulary through connector graphs and learns assembly sequences from human-designed LDraw objects [1]. BrickBench evaluates agentic text-conditioned design through validity, alignment, and design criteria [2]. These tasks motivate recognizable objects and executable checks; our agent instead receives a fixed object specification and must restore or complete it. We reuse the available brick geometry and connector infrastructure and do not claim a new collision or force simulator.

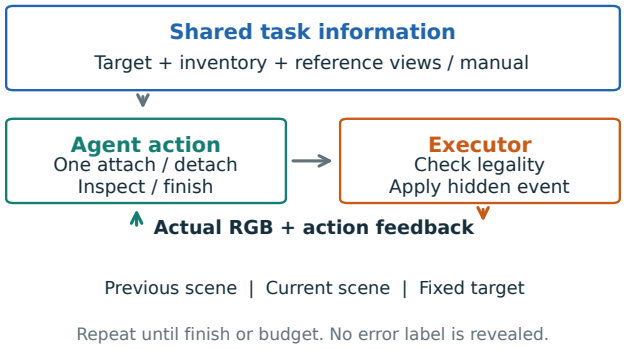


Figure 2. **The continuing interaction loop.** Target information stays available. Every manipulation is checked before the scene changes. The next observation reports the actual state, including any hidden event, without an error label or recovery command.

Break and Make evaluates interactive inspection, disassembly, and subsequent reconstruction [8]. LEGO-Puzzles studies elementary spatial reasoning and multistep planning, while PhyBlock evaluates physical understanding and block assembly planning [4, 6]. Our module task reduces ambiguity about unseen structure by supplying reference instructions, then introduces a change during the agent's own progress.

Manual execution and error recovery. MEPNet translates visual manuals into machine-executable assembly plans [10]. LEGO-ECA and SCANet pair assembly-step manual images with failures and corrections [9]. They are particularly close to our static task. Our dynamic condition retains a single agent's continuing history before and after an external change, with opportunities to inspect, repair, and finish.

AsgardBench evaluates visually grounded plan adaptation with lightweight feedback [7]; RoboRecover evaluates policies from reconstructed execution-deviation states [3]. Thus dynamic recovery itself is not new. The distinction here is the combination of instruction-assisted brick modules, one-part geometric execution, and an explicit test of how reference-order assumptions change which trajectories are evaluated.

3. Task and Execution Contract

3.1. State, observations, and actions

A task consists of an immutable context B, movable inventory P, candidate poses C_p for each part p, an initial state s_0, and an acceptance target g. The state maps each movable part to an installed pose or the rack: $s(p) \in C_p \cup \{\perp\}$. The context is already assembled and cannot be removed. In the static and dynamic object/mechanism tracks, the environment includes a white baseplate. The community module suite additionally preserves the original surrounding object as a rigid fixture. We do not equate completing that module with building the entire source object from an empty table.

At decision t, the module observation contains the fixed acceptance target, initial context, manual, inventory, the previous scene, and the actual current scene (Fig. 2). Three fixed views—isometric, rear, and top—remain available. The first observation explicitly lacks a previous action. Public part and placement handles are randomized; neither their names nor image filenames encode a goal label. An inspection can request another view, a part card, or a candidate-placement preview. A preview depicts a hypothetical placement and does not change the world or certify that placement as correct.

The action space is

$$a_t \in \{\text{attach}(p, c), \text{detach}(p), \text{inspect}(q), \text{finish}\}. \quad (1)$$

One manipulation moves one brick. The executor checks pose validity, endpoint collision, a compatible connection for attachment, and a collision-free straight insertion or extraction path. Rejected actions leave the state unchanged and return a legality code. These codes can assist recovery; they are recorded separately from visual observations. The module manipulation budget is $4|P_g|+12$, where P_g denotes the normal target pieces; inspections are counted separately. D instead uses a decision budget $\max(40, 6|P_g|+20)$ that includes inspections and invalid outputs. The static reference cost is a feasible repair witness, not a claim of globally shortest repair. Comparisons use the same track-specific limits for every evaluated model.

3.2. Completion without a prescribed trajectory

Let $\sigma(s)$ be the multiset of installed part geometries, colors, and poses, accounting for explicitly supported symmetries of plain rectangular pieces. Completion is

$$C(\tau) = \mathbf{1}[a_T = \text{finish}]\mathbf{1}[\sigma(s_T) = \sigma(g)]. \quad (2)$$

All preceding accepted actions must pass the executor. Swapping indistinguishable pieces does not change the terminal signature. Candidates include interchangeable placements for matching pieces. A floating brick, intersecting placement, duplicate attachment, or state replacement cannot be accepted as a shortcut.

A manual $M = (\bar{a}_1, \ldots, \bar{a}_n)$ supplies one verified build witness and images explaining each part's intended placement (Fig. 3). Its intermediate states $\bar{s}_k$ are not constraints on s_t. Figure 4 gives a concrete counterexample to equating equal progress with equal geometry. Static repair uses the same terminal rule, but its acceptance target is the explicit stage image supplied with that question, rather than the completed module. The stage index identifies that target image; it does not constrain the agent's repair sequence.

3.3. Hidden interventions on the actual state

After an accepted manipulation, let $\tilde{s}_{t+1}$ be the executor's result. In M, a condition specifies an error family f and minimum installed count k. The event selector searches for an edit e when at least k normal inventory pieces are installed. A valid edit must (i) change the actual geometry or color, (ii) consist of legal physical primitives, and (iii) have a legal inverse restoring the pre-event state. For removal and color replacement, it selects a previously installed part and retains the newest part. An addition uses a compatible exposed connection. Candidates are considered deterministically given the trajectory.

$$s_{t+1} = \begin{cases} e(\tilde{s}_{t+1}) & \text{if a feasible, unused event is selected,} \\ \tilde{s}_{t+1} & \text{otherwise.} \end{cases} \quad (3)$$

The selector neither compares $\tilde{s}_{t+1}$ with a reference prefix nor asks whether the agent installed a particular set of target pieces. If no feasible edit exists, the event is deferred. At most one event is applied in the current module suite. A color replacement comprises two legal environment primitives, detach then attach, executed between agent observations. The agent still receives only one physical move per action.

The inventory and candidate catalog are identical across an object's control and fault conditions. Each normal piece has a same-shape alternative-color spare, and one extra red plate is available, giving $2|P_g|+1$ inventory pieces. A different legal order can produce a different feasible intervention location. Accordingly, *completion scoring* is independent of the chosen reference order, while *intervention selection* is adaptive and need not yield identical fault difficulty across trajectories.

4. Benchmark Construction

4.1. Complete coverage and version continuity

The current benchmark has three tracks (Table 1), totaling 3,340 static and 1,920 dynamic case records. The 160 S configurations, 68 D structures, and twelve M modules are 240 *track-specific* configuration entries, not 240 independent physical models. Fifteen model identifiers occur in both S and D; all twelve M source objects also occur among



Figure 3. **A complete seven-part reference manual.** The seaplane's surrounding assembly is supplied. Each column shows one physical part and the state after placement. This is one executable route; alternative legal orders are accepted. Orange outlines identify the newly added part and are absent from ordinary current-state observations.

Table 1. **Dataset overview and task inputs.** Controls are included in case totals. Parts count the task's target pieces, excluding supplied context. M-static and M-dynamic share the same twelve modules. S has eleven non-control category labels, including replacement repair states; these are not eleven distinct error mechanisms. Sources overlap across tracks.

Track	Models	Cases	Controls	Parts	Errors	Initial state and visual guidance
S: static	160	3,292	160	7–16	11	Current scene → stage target; inventory and inspection.
D: dynamic	68	1,872	160	4–15	12	Empty board → full target; inventory and temporal views.
M: static	12	48	12	5–12	3	Supplied current stage → stage target; full remaining-part manual.
M: dynamic	12	48	12	5–12	3	Supplied context → completed module; manual and temporal views.

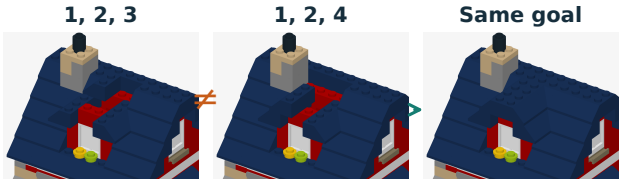


Figure 4. **Different valid prefixes.** Both roof prefixes contain three pieces. Installing reference piece 4 before piece 3 is legal and reaches the same target. Manual indices identify placements, not compulsory action times.

S's sources, often with different selected parts. This overlap is retained in provenance and considered in split design.

All 127 earlier static configurations remain in S, which adds 33 configurations. All 30 earlier dynamic structures remain in D; 38 new object designs expand its complete-object collection from four to 42. The chair, table, small car, and crane are therefore part of the present benchmark and its proposed experiments. Earlier case totals describe previous versions and are not added to the current 5,260 records.

M: instruction-guided community modules. We select twelve original LDraw OMR scenes and preserve all source instances, author metadata, original files, and source transforms. The task removes a small connected-to-context module from each scene; the surrounding object is supplied as a fixture. The original scenes contain 52–556 instances, including figures and scenery, while the remaining tasks contain only 5–12 parts, totaling 99 normal placements. A complete-object view establishes what is being built; fixed module views make the remaining placement and errors visible.

For each module we construct a no-error condition and three error families: one missing part, a same-shape color replacement, and an extra plate. Each is instantiated as static repair and dynamic completion, yielding $12 \times 4 \times 2 = 96$ cases. Static targets and fault snapshots are paired with the illustrated reference route's dynamic checkpoint. This pairing does not imply that every alternative route encounters the same affected brick. Figure 5 places module faults alongside earlier S and D examples. The supplement shows all 42 complete objects, all 26 mechanisms, and all twelve modules, and enumerates all 160 static configurations.

S: source-derived static repair. The white-baseplate bank contains 3,292 questions across 160 model configurations. Its migration ledger records 2,144 directly replayable plans, 834 newly verified dismantle/rebuild witnesses, 37 witnesses using held components, and ten corrected initial scenes that had penetrated the board. Another 267 incompatible source cases were replaced with distinct witnessed repair states, preserving replacement provenance. All 3,292 final witnesses replay under the new rules. This establishes geometric executability; the complete RGB package and per-question human readability review remain unfinished.

D: complete objects and structural mechanisms. The broader bank contains 1,872 episodes on 68 structures, including 42 recognizable simplified objects. There are 160 normal controls and 1,712 interventions in twelve families. These cover removals, shifts, rotations, swaps, color/size/type substitutions, added layers, early covers, buried color errors, cascaded removals, and compound faults. Complete-object tasks use 6–15 parts. Their templates and evaluation history are versioned separately from the new community modules; existing template locations do not acquire new visual validation merely because an adaptive event is executable.

Track-specific visual information. S supplies the current faulty scene, its acceptance target, inventory, and inspection tools. D begins from the empty white workbench and supplies a fixed complete target, inventory, and temporal observations; its current release does not supply M's complete step-by-step manuals. M begins from a supplied context and makes every remaining reference step accessible. Thus cross-track score differences cannot isolate object complexity or manual benefit. The planned manual ablation is within M, with identical objects and initial states. D's live event selector requires a certifiable continuation from a correct prefix; M selects feasible edits from its actual installed state. Both record deferred and untriggered events.

Error mechanisms and human assembly analogues. Figure 5 combines examples from every track. Missing pieces model forgotten or detached work; shifts and rotations model incorrect alignment; color, size, and type substitutions model selection mistakes. An added layer introduces an unintended extra part, while an early cover can obstruct later access. A buried defect requires undoing correct work, a cascade removes several connected pieces, and a compound event combines mistakes. These are geometric analogues of assembly errors, not measured frequencies of human behavior. D contains all twelve mechanisms; M currently contains missing, color-replacement, and extra-plate events.

4.2. Geometry and visibility admission

Every operated part must have usable collision geometry. Each installation, intervention primitive, and reference recovery is checked against the present assembly and environment. Unsupported background collision meshes in community scenes are retained visually and represented by conservative bounding boxes for endpoint and swept-path rejection; they are never silently omitted as obstacles. The context's internal construction history, force stability, and grasp mechanics are outside the task's claims.

The module renderer uses the same camera and framing for current, reference, and fault states. In each module, every manual addition has at least 128 visible pixels in one of the three 900-by-660 interface views; each reference fault changes at least 128 pixels by at least 24 RGB intensity levels in one view. These are reproducible screening thresholds, not perceptual guarantees. They apply to the checked reference scenes; an adaptive event at another location needs its own visibility record before supporting a visual-miss claim.

The mechanism bank separately records 1,553 primary temporal candidates, 136 events that cancel the latest action, and 23 low-contrast cases. Cancellation matters: if a new part is immediately removed, the scene before the action and the next current scene may look identical. That tests action-outcome monitoring and should not be silently pooled with changes to older completed work.

5. Evaluation and Diagnostic Experiments

5.1. Completion, exposure, and recovery

For a scheduled fault trial i, let T_i indicate an actual intervention and C_i the completion criterion in Eq. 2. We report

$$A_{\text{complete}} = \frac{1}{N}\sum_i C_i, \qquad A_{\text{trigger}} = \frac{1}{N}\sum_i T_i, \quad (4)$$

$$A_{\text{recover}} = \frac{\sum_i T_i C_i}{\sum_i T_i}. \quad (5)$$

An untriggered but correctly completed episode counts as completion, not as a successful repair or a completion failure. Empty conditional denominators remain undefined. A control false alarm, invalid output, rejected manipulation, or service failure is recorded as its own outcome. Inspection count and physical action count are also reported; the stored recovery witness is feasible and is not claimed to be globally optimal.

A final correct object establishes effective recovery after exposure, but does not by itself establish visual detection. A detection study requires a separate preregistered judgment or a trace criterion with clear timing relative to legality feedback. We therefore avoid labeling every incomplete episode a visual miss. To compare recovery causally, a paired evaluation branches a saved agent/environment prefix into control and intervention continuations, retaining the same preceding observations and actions. Independently generated normal trajectories are useful construction controls but are not matched-prefix evidence.

Aggregation and splits. Case variants from one object are dependent. Model comparisons should report per-source and per-error results, source-macro averages, and uncertainty intervals clustered by source. The mechanism release has a 43-family development/evaluation split; its balanced subsets contain 108 and 168 intervention cases, respectively. The community suite is currently a development

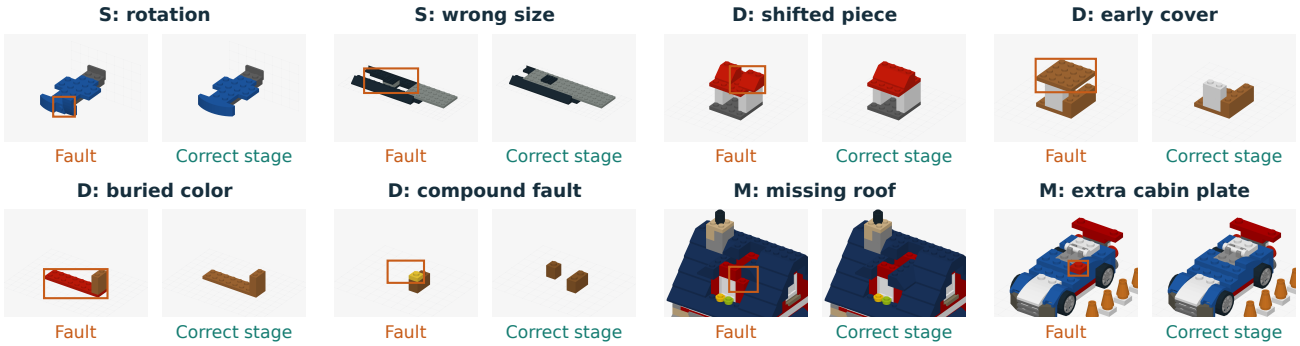


Figure 5. **Eight compact fault/target pairs across all tracks.** S: rotation and size; D: shift, premature cover, buried color, and compound error; M: missing roof and extra plate. Within each cell the left image is faulty and the right is the correct stage, using identical framing. Orange boxes explain the figure only. The supplement illustrates all twelve D mechanisms and eight S categories.

and protocol-diagnostic collection, not an unseen-source leaderboard. Overlap between older static and dynamic sources precludes claiming cross-bank generalization from their names alone. The supplementary protocol specifies the additional experiments needed for a learned-agent comparison.

5.2. Does the trigger depend on one build order?

We run a deterministic diagnostic over all twelve community modules. For each, the reference route and one physically verified alternative route are executed under control, missing-part, wrong-color, and extra-part conditions. This gives 96 dynamic route–condition trials: 24 controls and 72 fault trials. Alternative routes are deliberately selected as regression tests, favoring a different physical state at the intervention threshold when feasible. They are not random samples from the set of all legal assembly orders.

We compare an *exact-prefix trigger*, which requires equality with the reference checkpoint, against the actual-state selector. To measure the former's opportunity without introducing a different repair policy, we inspect each otherwise normal route for that reference state. Both schedules cover all 36 reference-route fault trials. On the alternative routes, exact-prefix eligibility falls to 9/36, whereas actual-state interventions occur in 36/36 (Table 2). The difference comes from nine objects whose threshold states differ physically from the reference. Thus the diagnostic isolates a protocol effect with identical target objects and legal construction routes; it does not measure a language model's tendency to choose those routes.

The corresponding intermediate-state acceptance rule rejects all twelve alternative normal routes, despite their correct endpoints. Terminal multiset comparison accepts them all. Separate regression checks exchange identical part identities in each of the twelve modules and preserve success, while rejecting repeated, floating, and overlapping attachments and premature completion. These checks exercise both sides of the contract: freedom to choose a valid sequence does not permit fictitious physical actions.

Table 2. **Route sensitivity diagnostic.** All 24 normal routes are legal and reach their targets. Fault-trial denominators are three families per object. These are deterministic executor results.

Check	Reference	Alternative
Normal terminal geometry correct	12/12	12/12
All intermediate states match manual	12/12	0/12
Exact-prefix trigger opportunity	36/36	9/36
Actual-state intervention triggered	36/36	36/36

5.3. Are interventions consequential and recoverable?

A blind continuation policy attempts the reference or alternative normal route without reading the changed state. A privileged recovery policy uses the evaluator's feasible inverse after a triggered edit, then continues that route. The blind policy completes all 24 controls and none of the 72 fault trials; privileged recovery completes all 96 trials. These controls show that the selected faults alter the endpoint and that a legal recovery exists. Neither policy tests visual understanding. The supplement illustrates certified recoveries in the racer and table.

The broader mechanism bank provides complementary checks: blind continuation completes 160/160 controls and 0/1,712 intervention episodes; a privileged reset/rebuild witness completes 1,872/1,872. We report these within their own bank because event templates, initial states, and action histories differ from those of the module suite.

Legality versus correctness in actual trajectories. Figure 6 distinguishes external edits, legal but incorrect placements, and valid alternative orders. All 32 decisions in the two archived model runs reproduce their saved state hashes

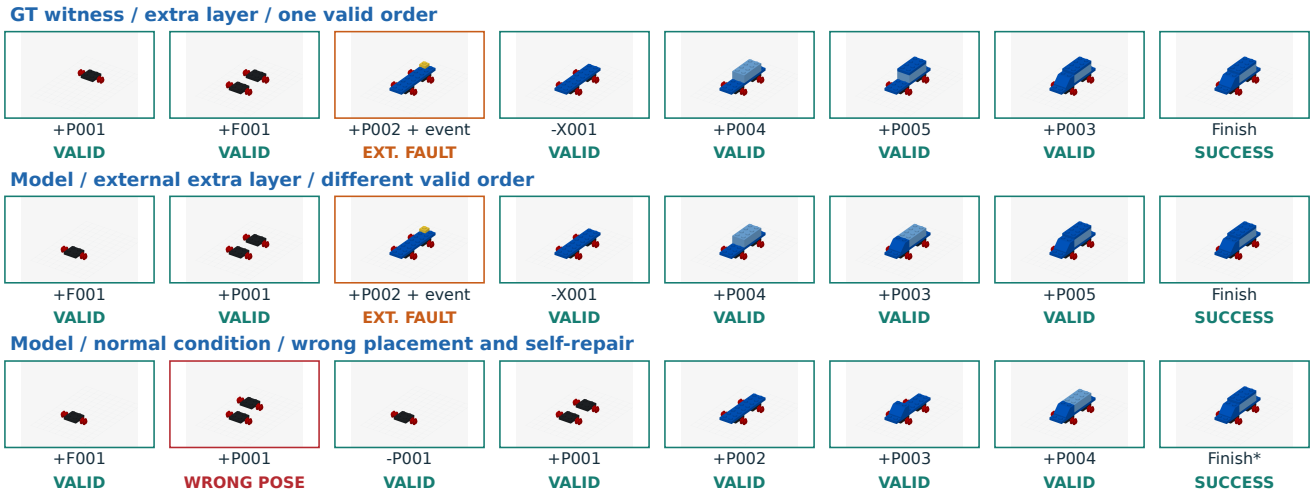


Figure 6. **GT versus actual model repair, eight states per row.** Top: one legal reference witness (GT) for an extra-layer event. Middle: an archived Gemini 4 Argon rollout removes the extra piece and succeeds with a different legal order. Bottom: a separate normal-condition rollout legally places P001 at the wrong pose, removes it, and corrects it. Green: valid partial state or terminal success; orange: external intervention; red: wrong placement. All three finish correctly. +/− denote attach/detach; selected views omit inspections and intervening actions (*). These replay-verified v5 examples are development evidence, not current-suite scores or matched-prefix causal pairs.

Table 3. **Diagnostic endpoints.** Success counts from replayed scripts. Privileged policies access hidden state and are reachability checks, not visual-agent baselines.

Bank	Policy / condition	Completed
Modules	Blind / normal	24/24
Modules	Blind / fault	0/72
Modules	Privileged / all	96/96
Mechanisms	Blind / normal	160/160
Mechanisms	Blind / fault	0/1,712
Mechanisms	Privileged / all	1,872/1,872

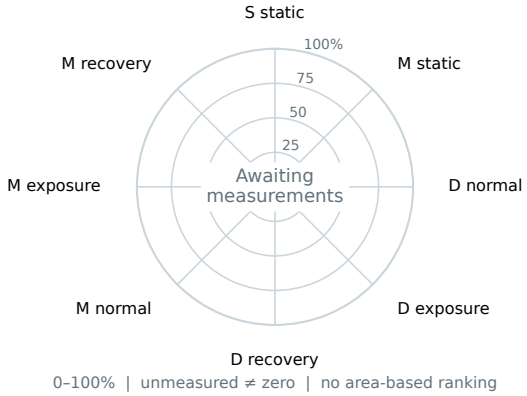


Figure 7. **Evaluation profile template.** Eight axes correspond to Table 4: static success, normal construction, intervention exposure, and conditional recovery. All axes use 0–100%; no model curve is drawn before measurement. Exposure measures protocol coverage, not detection accuracy. Counts and uncertainty remain in tables; polygon area is not an overall score.

and feedback. The red intermediate state is a corrected mistake, not a failed episode.

5.4. Planned model and human evaluation

Table 4 reserves the primary model comparisons. Every dash denotes an uncollected measurement. We propose five visual models: three commercial families and two open-weight models of different sizes. Before execution, record exact provider/checkpoint identifiers, revision dates, prompts, image preprocessing, and decoding settings. Use three seeds (41–43), identical information and action budgets within a track, and report source-macro means with source-clustered 95% intervals. The artifact supplies the case manifest; the supplement provides a human comparison panel; no paid evaluation is performed for this design.

Primary and coverage runs. S uses 467 test questions from 20 source groups after visual admission; current-source reassignment corrects 151 stale migrated split entries. D uses 168 faults (14 per mechanism) and 72 normal definitions. Each fault receives cloned control/fault continuations; the 72 normal definitions are additional controls. M uses 48 static cases, 36 dynamic fault pairs, and twelve normal episodes, without an unseen-source claim. Full-bank runs are separate from these primary tables. Cross-track source overlap must be checked before joint tuning or transfer studies.

Table 4. **Planned primary model results—not yet measured.** S: 467 static test questions; M-static: 48 public cases; D: 168 fault pairs + 72 normal definitions; M-dynamic: 36 fault pairs + 12 normal episodes. C: normal completion; T: actual intervention coverage; R: recovery conditional on exposure (Eq. 5). Values will be percentages with source-clustered intervals. "—" means unmeasured, never zero.

	Static success		D: objects/mechanisms			M: modules		
Model slot	S	M	C	T	R	C	T	R
V1: commercial family A	—	—	—	—	—	—	—	—
V2: commercial family B	—	—	—	—	—	—	—	—
V3: commercial family C	—	—	—	—	—	—	—	—
V4: open-weight, smaller	—	—	—	—	—	—	—	—
V5: open-weight, larger	—	—	—	—	—	—	—	—

Table 5. **Planned input ablations.** Unmeasured template, instantiated separately for V1 and V5; geometry, seeds, and cases are paired. R: exposure-conditioned recovery; I: illegal-attempt rate.

Observation / feedback	D R	M R	I
Full visual interface	—	—	—
No previous-state image	—	—	—
No manual (M only)	n/a	—	—
No candidate preview	—	—	—
Success/failure feedback only	—	—	—
Symbolic state (privileged)	—	—	—

Human operation and input diagnosis. Use twelve control/fault pairs each from S, D, and M (72 conditions). Six participants complete 36 conditions each; counterbalancing gives three observations per condition without exposing both pair members to one participant. All five models use this common panel. Humans select visual actions without written explanations or JSON. Record completion, exposure, recovery, false alarms, actions, and time. This baseline is not a matched-history causal estimate.

Ablations and failure analysis. Table 5 reserves within-track input interventions for one commercial and one open-weight model chosen before evaluation. Remove previous-state images, candidate previews, or detailed legality feedback while retaining task geometry. Remove the manual only in M; S and D have no equivalent full manual to remove. A symbolic-state condition diagnoses the visual bottleneck and remains privileged. Report errors separately as invalid output, rejected physical action, untriggered event, incomplete planning, and valid but wrong anomaly judgment; final failure alone is not a detection miss. Break down D's twelve mechanisms, part-count bands, event stage, and temporary-disassembly requirements. Empty strata and conditional denominators remain undefined. Figure 7 specifies the aggregate profile; the supplement provides a second radar for recovery across the twelve D error families.

Efficiency and paired recovery. Record decisions, inspections, accepted/rejected moves, latency, tokens, and charges, including both continuations and shared prefixes. Also report recovery among exposed pairs whose normal branch succeeds, with its denominator. Keep service failures and prefix stalls in the ledger; budget from a current-interface pilot, not archived car costs.

6. Discussion and Limitations

The route study establishes a protocol counterexample, and geometry checks establish witnessed completions; neither ranks learned agents. Formal model results, independent human operation, and complete static visual admission remain pending. Archived car traces and binary human judgments are development evidence only (supplement).

Tracks share sources and expose different information; combined case counts are not independent sample sizes, and cross-track differences cannot isolate perception or difficulty. Pixel thresholds do not ensure human readability, and adaptive event locations need new visibility checks. The executor models rigid connectors and straight motion with finite pose catalogs, not force stability, finger clearance, deformation, or robot execution. Source attribution accompanies the artifact.

7. Conclusion

BRICKPLAN brings the complete static and dynamic collection into a common, source-linked evaluation framework while retaining each track's input contract. Its 5,260 case records cover earlier objects and mechanisms as well as the new community modules. Terminal geometric scoring permits alternative legal routes, and actual-state interventions preserve testing opportunities that reference-prefix triggers can miss. The executable diagnostics and explicit model/human experiment design support a reproducible study of visual assembly, error monitoring, and recovery.

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